

JKBOSE 2026

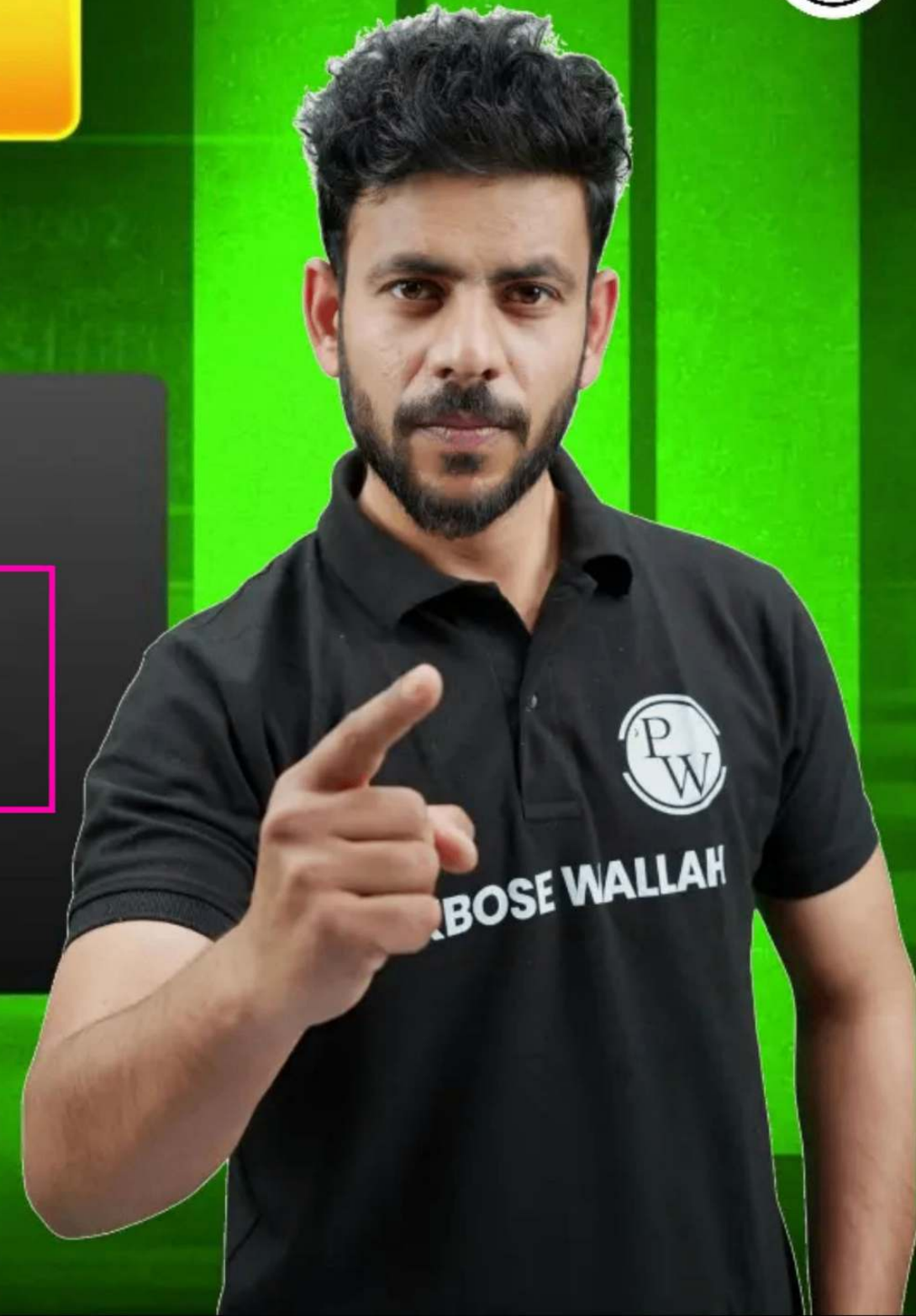
CLASS 11TH

THE LIVING WORLD

LECTURE-04

BOTANY

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Topics To Covered

- 1** Taxonomic Categories ✓
- 2** Taxonomic Hierarchy ✓
- 3** Species ✓
- 4** Genus ✓

Common Names

1. Potato
2. Brinjal
3. Makoi / Black night Shade
4. Lion
5. Tiger
6. leopard
7. Dog
8. wolf
9. Cat
10. human
11. Housefly

Scientific names

Solanum tuberosum
Solanum melongena
Solanum nigrum
Panthera leo
Panthera tigris
Panthera pardus
Canis familiaris
Canis lupus
Felis catus
Homo sapiens
Musca domestica

Common names

12. Mango

13. Wheat

14. Rice

15. Maize

Scientific names

Mangifera indica

Triticum aestivum

Oryza sativa

Zea mays.



TAXONOMIC CATEGORIES

- Classification is not a single step process.
- It is a multistep process which involves hierarchy of steps in which each step represents a rank / category.

(*) Taxon = It is a unit of classification
(plural = taxa)

imp Note: There is thin line of difference b/w taxon and Category.

<u>Taxon</u>		<u>Category</u>
→ It represents living organism.	→	It represents rank of an organism.
→ Ex. <u>Solanum</u>	→	Ex. Genus
<u>tuberosum</u>	→	Species

(*)

<u>Solanum</u>	<u>tuberosum</u> (potato)
↓	↓
<u>Genus</u>	Specific Epithet.

④ There are 07 main/obligate taxonomic categories which are used necessary during any classification.

① Kingdom , ② Phylum / Division , ③ Class , ④ Order , ⑤ Family , ⑥ Genus , ⑦ Species.

Kingdom (Animal classification) ↓ Highest taxonomic category.

Phylum / Division (plant classification) ↓

Species ↓ Smallest taxonomic category.

* ④ Trick = Keep Pots Clean Or Family Gets Sick

All taxonomic Categories are distinct biological entities and not merely morphological aggregates.

صرف

1.2 TAXONOMIC CATEGORIES

Classification is not a single step process but involves hierarchy of steps in which each step represents a rank or category. Since the category is a part of overall taxonomic arrangement, it is called the **taxonomic category** and all categories together constitute the **taxonomic hierarchy**. Each category, referred to as a unit of classification, in fact, represents a rank and is commonly termed as **taxon** (pl.: taxa).

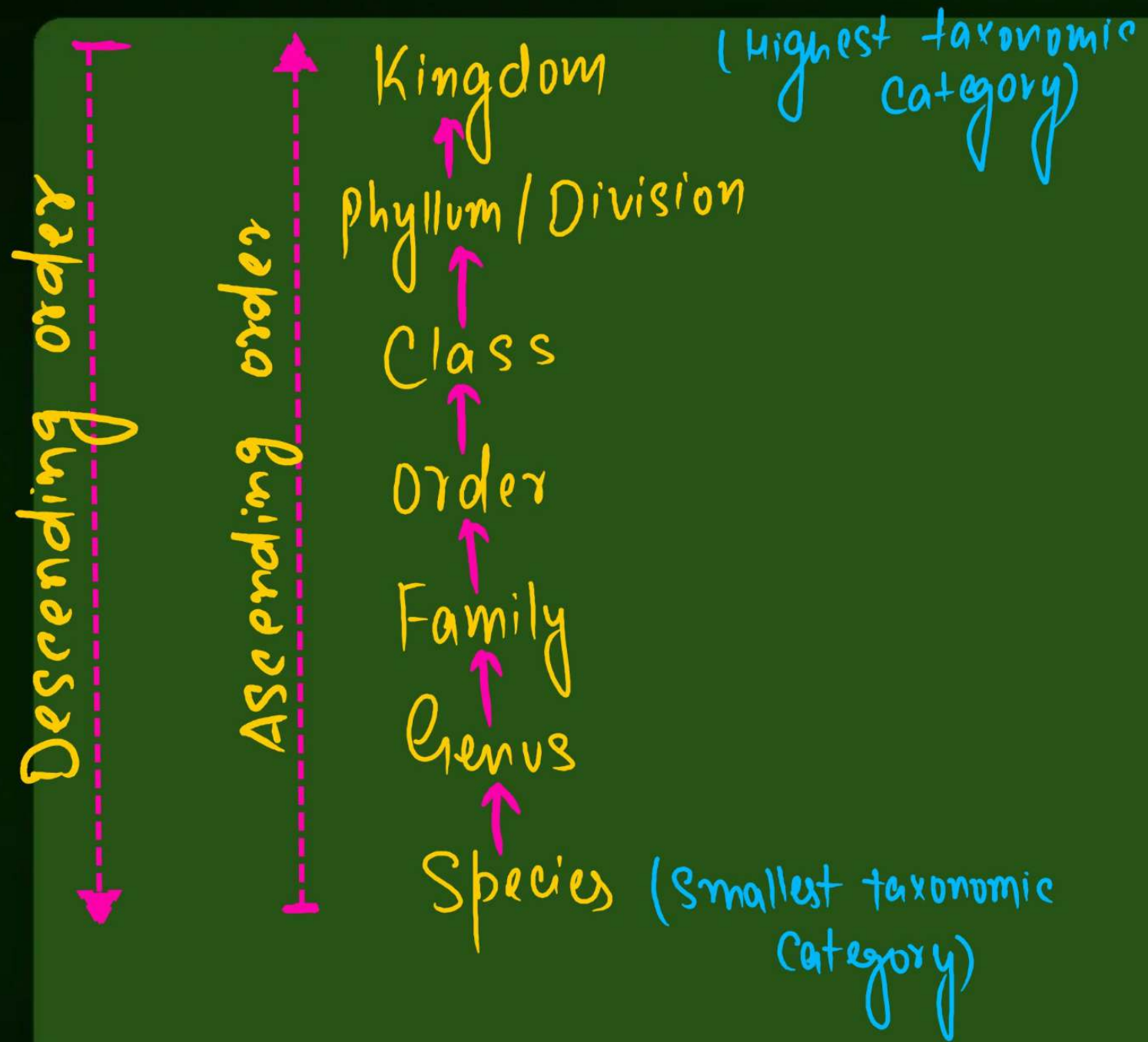
Taxonomic categories and hierarchy can be illustrated by an example. Insects represent a group of organisms sharing common features like three pairs of jointed legs. It means insects are recognisable concrete objects which can be classified, and thus were given a rank or category. Can you name other such groups of organisms? Remember, groups represent category. Category further denotes rank. Each rank or *taxon*, in fact, represents a unit of classification. These taxonomic groups/categories are distinct biological entities and not merely morphological aggregates.

Taxonomical studies of all known organisms have led to the development of common categories such as kingdom, phylum or division (for plants), class, order, family, genus and species. All organisms, including those in the plant and animal kingdoms have species as the lowest category. Now the question you may ask is, how to place an organism in various categories? The basic requirement is the knowledge of characters of an individual or group of organisms. This helps in identifying similarities and dissimilarities among the individuals of the same kind of organisms as well as of other kinds of organisms.

TAXONOMIC HIERARCHY

Hierarchy = arrangement of things either in ascending order / descending order.

(*) Taxonomic hierarchy = arrangement of taxonomic categories
either in ascending order or
descending order.



*) AS we move from Sps. to Kingdom:

- a. No. of similar characters (i.e. specificity) decreases.
- b. No. of general characters increases.
- c. Complexity of classification increases because no. of general characters increases.



Figure 1.1 Taxonomic categories showing hierarchical arrangement in ascending order

The taxonomic categories from species to kingdom have been shown in ascending order starting with species in Figure 1.1. These are broad categories. However, taxonomists have also developed sub-categories in this hierarchy to facilitate more sound and scientific placement of various taxa.

Look at the hierarchy in Figure 1.1. Can you recall the basis of arrangement? Say, for example, as we go higher from species to kingdom, the number of common characteristics goes on decreasing. Lower the taxa, more are the characteristics that the members within the taxon share. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level. Hence, the problem of classification becomes more complex.

QUESTIONS

ICBN is:

[2007]

- 1 International Code for Biological Naming
- ✓ 2 International Code for Botanical Nomenclature
- 3 International Class for Biological Nomenclature
- 4 International Classification for Biological Nomenclature

QUESTIONS

In the taxonomic categories which hierarchical arrangement in ascending order is correct in case of animals? [MR* 2022]

Phylum ✓

NEET

- 1 Kingdom, Order, Phylum, Class, Family, Genus, Species
- 2 Kingdom, Phylum, Class, Order, Family, Genus, Species
- 3 Kingdom, Class, Phylum, Family, Order, Genus, Species
- 4 Kingdom, Order, Class, Phylum, Family, Genus, Species

language of Question
does not match c
answers.

QUESTIONS

NEET
[1999]

Which arrangement is in correct ascending order?

- 1 Species < Genus < ~~Order~~ < Family
- 2 ~~Genus~~ < Species < Family < Order
- 3 ~~Order~~ < Family < Genus < Species
- 4 Species < Genus < Family < Order

Kingdom
Phylum / Division
Class
order ✓
Family ✓
Genus ✓
Sps ✓

QUESTIONS

'Taxon' is the unit of a group of:

[1996]

- 1 Order
- 2 Taxonomy
- 3 Species
- 4 Genus

"Home work"

Thank
You